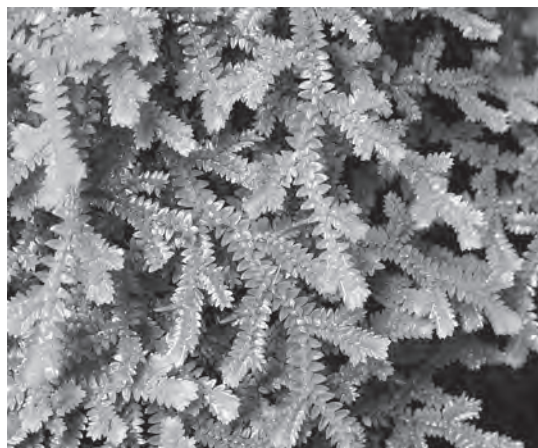


- 2 The photograph shows a type of plant called a liverwort. This plant grows in Costa Rica.

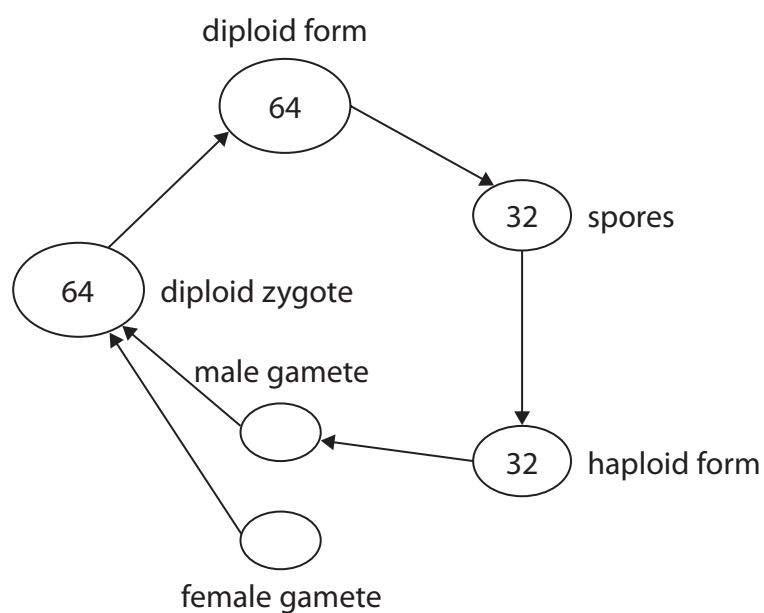


(Source: © Steve Taylor ARPS/Alamy Stock Photo)

- (a) Liverworts reproduce sexually with gametes and asexually with spores.

The diagram shows the life cycle of a liverwort, which exists in two forms.

The diploid form contains 64 chromosomes and the haploid form contains 32 chromosomes.



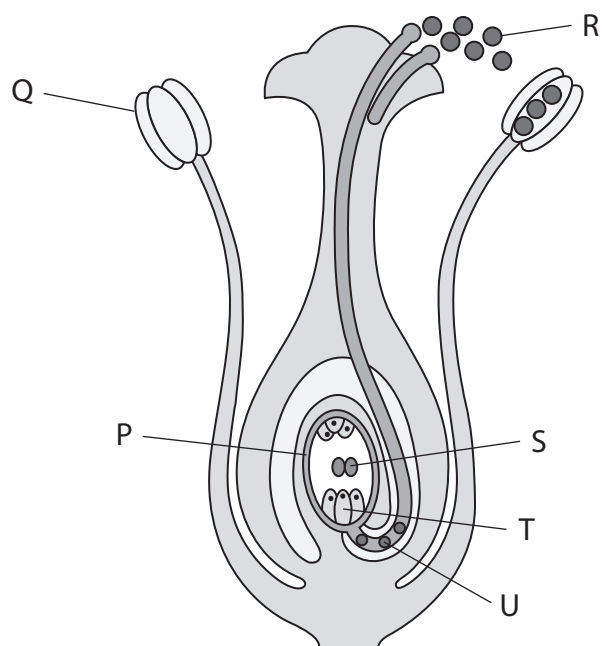
- (i) Complete the diagram to show the number of chromosomes in the female and male gametes.

(1)

- (ii) Label the part of the diagram where meiosis would have occurred.

(1)

(b) The diagram shows the nuclei in an ovule and a pollen tube in a flowering plant.



(i) Which letter labels an ovule?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** S
- ☐ **D** T

(ii) Which letter labels an egg cell (female gamete)?

(1)

- ☐ **A** P
- ☐ **B** S
- ☐ **C** T
- ☐ **D** U

(iii) Which letter labels a male nucleus?

(1)

- ☐ **A** R
- ☐ **B** S
- ☐ **C** T
- ☐ **D** U



